

CS 305 Lab Tutorial

Lab 12

communication and learning in network

Layer3: Router and Routing Protocol

Layer2: Switch and ARP(for IPv4) vs NS&NA(for IPv6)

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Topic

- Routing Protocol : RIP vs OSPF
 - RIP (RIP1, RIP2 for IPv4, RIPng for IPv6)
 - OSPF (OSPFv1, OSPFv2 for IPv4, OSPFv3 for IPv6)
- ARP for IPv4 vs NS&NA for IPv6
 - MAC address (Layer 2 address)
 - ARP request, ARP reply for IPv4
 - Neighbor Solicitation, Neighbor Advertisement in ICMPv6 for IPv6
- Device
 - Hub (broadcast) vs Switch (Active learning, mac-address table)
- Practice & Tips
 - Build network on simulator
 - Configure
 - Test

Tips on eNSP : useful commands while in CLI

➤ **display**

- **dis this**
- **dis ip routing-table**
- **dis ospf routing**
- **dis mac-address**

➤ **enable IPv6**

- **[Huawei] ipv6**
- **[interface type id-list] ipv6 enable**

➤ **configure interface**

- **ip address** [IPv4-address in dotted decimal] [mask]
- **ip address** [IPv6-address] [mask]

➤ **configure RIP**

- **rip** <process_id>
- **version** <version_id> <mask> <next-hop>
- **network** <network_id>

➤ **configure OSPF**

- **ospf** <process_number>
- **area** <area_id>
- **network** <network ID> <wildcast mask>

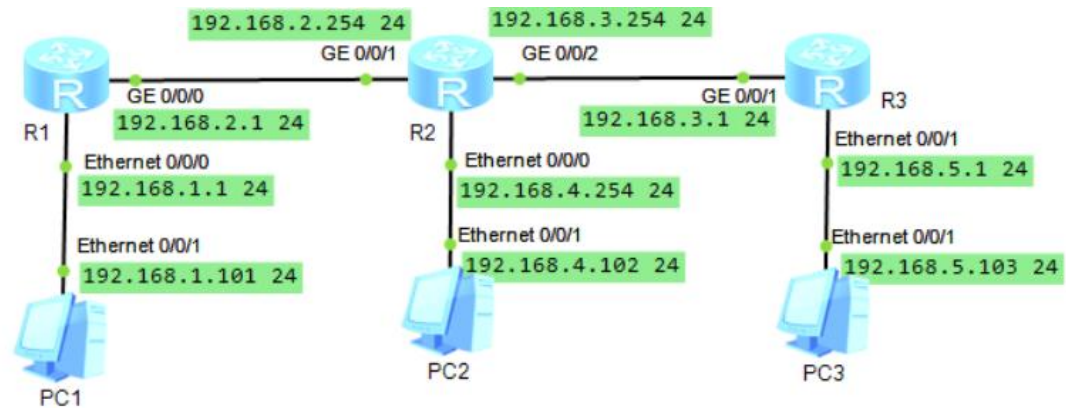
Part A. Routing Protocol (1)

RIP - Routing Information Protocol

- **Distance Vector** or Ford-Fulkerson algorithms
- This protocol is most useful as an "interior gateway protocol" (IGP)
- represents the metric as a sum of "costs" for individual hops
- update:
 - every gateway that participates in routing sends an update message to all its neighbors once every **30** seconds
 - wait for **180** seconds before timing out a route
- version:
 - RIP V1 (IPv4, broadcast, No authentication, ...)
 - RIP V2 (IPv4, multicast, authentication, ...)
 - RIPng (IPv6, multicast, authentication, ...)

RIP for IPv4(1)

- Configuration on eNSP
 - 0. interface configuration on the interface of Router
 - command: **rip 1**
 - 1. start rip configuration
 - command: **version** <version number>
 - 2. specify the rip version
 - command: **network** <network ID>
 - 3. add the network info which is connect directly by the Router
 - command: **network** <network ID>



```
R1
[Huawei] rip 1
[Huawei-rip-1] version 2
[Huawei-rip-1]
Dec 2 2023 10:29:07-08:00 Huawei DS/4/DA
.25.191.3.1 configurations have been chan
e change loop count is 0 and the maximum
[Huawei-rip-1] network 192.168.1.0
[Huawei-rip-1] network 192.168.2.0
```

The configurations about RIP in this demo are as following:

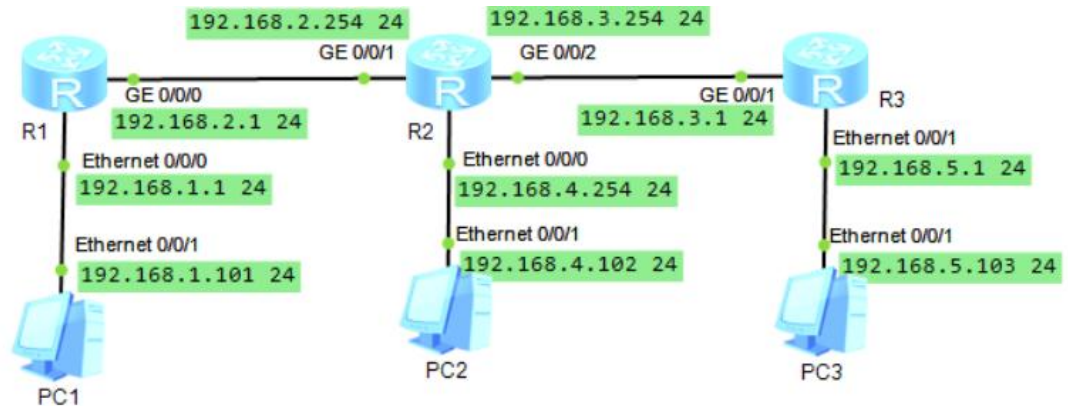
```
R1
[Huawei-rip-1]dis this
#
rip 1
version 2
network 192.168.1.0
network 192.168.2.0
#
return
```

```
R2
[Huawei-rip-1]dis this
#
rip 1
version 2
network 192.168.2.0
network 192.168.4.0
network 192.168.3.0
#
return
```

```
R3
[Huawei-rip-1]dis this
#
rip 1
version 2
network 192.168.3.0
network 192.168.5.0
#
return
[Huawei-rip-1]
```

RIP for IPv4(2)

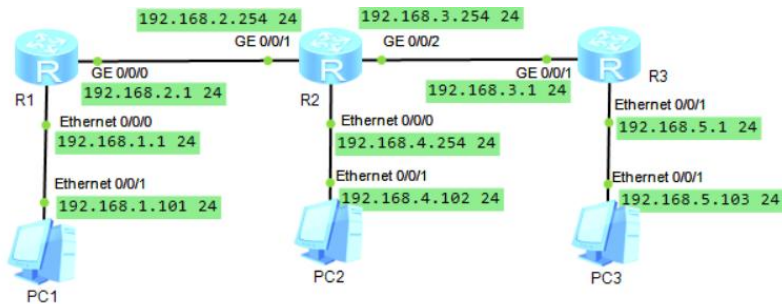
- After the configuration on RIP is done, the routing-table of R1 is as following:
 - the routing to 192.168.3.0/24, 192.168.4.0/24, 192.168.5.0/24 are generated by RIP
 - Cost is determined by the number of routers on the routing path



R1						
[Huawei]dis ip routing-table						
Route Flags: R - relay, D - download to fib						

Routing Tables: Public						
Destinations : 9 Routes : 9						
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	Direct	0	0	D	192.168.1.1	Ethernet0/0/0
192.168.1.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0
192.168.2.0/24	Direct	0	0	D	192.168.2.1	GigabitEthernet
0/0/0						
192.168.2.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
192.168.3.0/24	RIP	100	1	D	192.168.2.254	GigabitEthernet
0/0/0						
192.168.4.0/24	RIP	100	1	D	192.168.2.254	GigabitEthernet
0/0/0						
192.168.5.0/24	RIP	100	2	D	192.168.2.254	GigabitEthernet
0/0/0						

RIP for IPv4(3)



the routing-table of R2:

----->

the routing-table of R3:

----->

R2

[Huawei]dis ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 10 Routes : 10

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	RIP	100	1	D	192.168.2.1	GigabitEthernet
0/0/1						
192.168.2.0/24	Direct	0	0	D	192.168.2.254	GigabitEthernet
0/0/1						
192.168.2.254/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
192.168.3.0/24	Direct	0	0	D	192.168.3.254	GigabitEthernet
0/0/2						
192.168.3.254/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						
192.168.4.0/24	Direct	0	0	D	192.168.4.254	Ethernet0/0/0
192.168.4.254/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0
192.168.5.0/24	RIP	100	1	D	192.168.3.1	GigabitEthernet
0/0/2						

R3

[Huawei]dis ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 9 Routes : 9

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	RIP	100	2	D	192.168.3.254	GigabitEthernet
0/0/1						
192.168.2.0/24	RIP	100	1	D	192.168.3.254	GigabitEthernet
0/0/1						
192.168.3.0/24	Direct	0	0	D	192.168.3.1	GigabitEthernet
0/0/1						
192.168.3.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
192.168.4.0/24	RIP	100	1	D	192.168.3.254	GigabitEthernet
0/0/1						
192.168.5.0/24	Direct	0	0	D	192.168.5.1	Ethernet0/0/1
192.168.5.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/1

Part A. Routing Protocol (2)

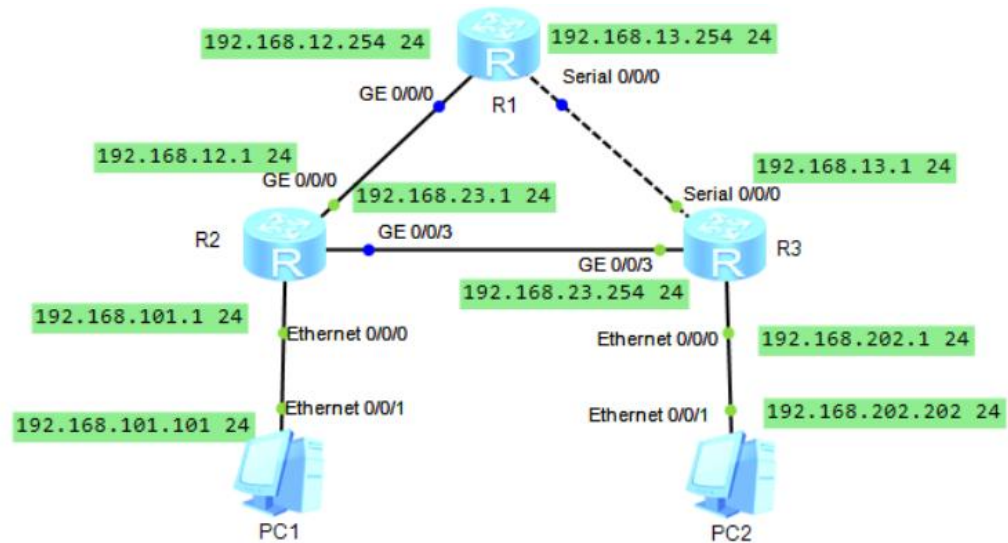
OSPF – Open Shortest Path First

- A **link-state** routing protocol
- OSPF is classified as an Interior Gateway Protocol (IGP)
- Each router maintains a database describing the Autonomous System's topology
- Version
 - OSPFv1, OSPFv2 for IPv4
 - OSPFv3 for IPv6

OSPF(1)

Configuration on eNSP

- step1 : start OSPF configuration
 - command: **ospf** <process number>
- step2: specify the index of area
 - command: **area** <area index>
- step3: add the network info which is connect directly by the Router
 - command: **network** <network ID> <wildcast mask>
 - **wildcast mask**: the result of performing bitwise inversion on a 32-bit subnet mask.
 - e.g. the subnet mask is: 255.255.255.0, the the wildcast mask is 0.0.0.255



R1

```
[Huawei]ospf 1
[Huawei-ospf-1]area 0
[Huawei-ospf-1-area-0.0.0.0]network 192.168.12.0 0.0.0.255
[Huawei-ospf-1-area-0.0.0.0]network 192.168.13.0 0.0.0.255
[Huawei-ospf-1-area-0.0.0.0]q
[Huawei-ospf-1]q
[Huawei]
```

R2

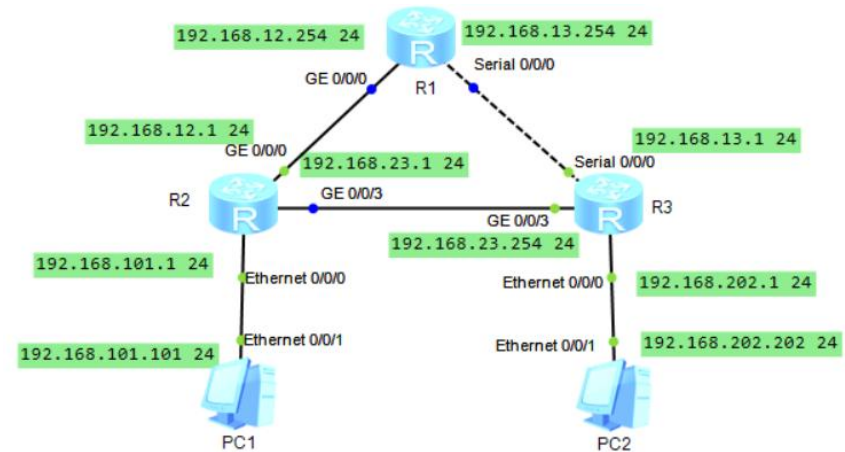
```
[Huawei]ospf 1
[Huawei-ospf-1]area 0
[Huawei-ospf-1-area-0.0.0.0]network 192.168.101.0 0.0.0.255
[Huawei-ospf-1-area-0.0.0.0]network 192.168.12.0 0.0.0.255
[Huawei-ospf-1-area-0.0.0.0]network 192.168.23.0 0.0.0.255
[Huawei-ospf-1-area-0.0.0.0]q
[Huawei-ospf-1]
```

R3

```
[Huawei]ospf 1
[Huawei-ospf-1]area 0
[Huawei-ospf-1-area-0.0.0.0]network 192.168.13.0 0.0.0.255
[Huawei-ospf-1-area-0.0.0.0]network 192.168.23.0 0.0.0.255
[Huawei-ospf-1-area-0.0.0.0]network 192.168.202.0 0.0.0.255
[Huawei-ospf-1-area-0.0.0.0]q
[Huawei-ospf-1]q
[Huawei]
```

OSPF(2)

After the configuration on OSPF is done, using command “**display ospf peer**” to view information about the router's neighbors



```
R1
[Huawei]dis ospf peer

      OSPF Process 1 with Router ID 192.168.12.254
      Neighbors

Area 0.0.0.0 interface 192.168.12.254(GigabitEthernet0/0/0)'s neighbors
Router ID: 192.168.101.1 Address: 192.168.12.1
State: Full Mode:Nbr is Master Priority: 1
DR: 192.168.12.1 BDR: 192.168.12.254 MTU: 0
Dead timer due in 29 sec
Retrans timer interval: 5
Neighbor is up for 00:17:48
Authentication Sequence: [ 0 ]

      Neighbors

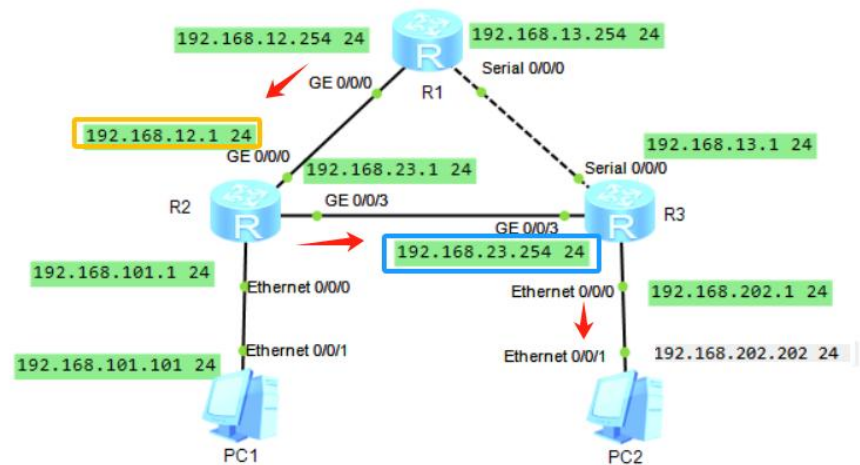
Area 0.0.0.0 interface 192.168.13.254(Serial0/0/0)'s neighbors
Router ID: 192.168.202.1 Address: 192.168.13.1
State: Full Mode:Nbr is Master Priority: 1
DR: None BDR: None MTU: 0
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00:15:21
Authentication Sequence: [ 0 ]
```

OSPF(3)

Initiate a “tracert” test from R1 to PC2, tracking the transmission path of the test packets

Q. Why R1 choose R2(192.168.12.1) other than R3(192.168.13.1) as the next hop ?

It seems that from R1 to PC2, passing directly through R3 is the least cost option?



```

R1
[Huawei] tracert 192.168.202.202

tracert to 192.168.202.202 (192.168.202.202), max hops: 30 ,packet length: 40,press CTRL_C to break
 1 192.168.12.1 50 ms 50 ms 40 ms
 2 192.168.23.254 60 ms 70 ms 60 ms
 3 192.168.202.202 90 ms 80 ms 100 ms
[Huawei]
    
```

```

R1
[Huawei]dis ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 10      Routes : 10

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
-----
127.0.0.0/8         Direct   0    0                D  127.0.0.1         InLoopBack0
127.0.0.1/32        Direct   0    0                D  127.0.0.1         InLoopBack0
192.168.12.0/24      Direct   0    0                D  192.168.12.254    GigabitEthernet
0/0/0
192.168.12.254/32    Direct   0    0                D  127.0.0.1         GigabitEthernet
0/0/0
192.168.13.0/24      Direct   0    0                D  192.168.13.254    Serial0/0/0
192.168.13.1/32      Direct   0    0                D  192.168.13.1      Serial0/0/0
192.168.13.254/32    Direct   0    0                D  127.0.0.1         Serial0/0/0
192.168.23.0/24      OSPF     10    2                D  192.168.12.1      GigabitEthernet
0/0/0
192.168.101.0/24     OSPF     10    2                D  192.168.12.1      GigabitEthernet
0/0/0
192.168.202.0/24     OSPF     10    3                D  192.168.12.1      GigabitEthernet
0/0/0
    
```

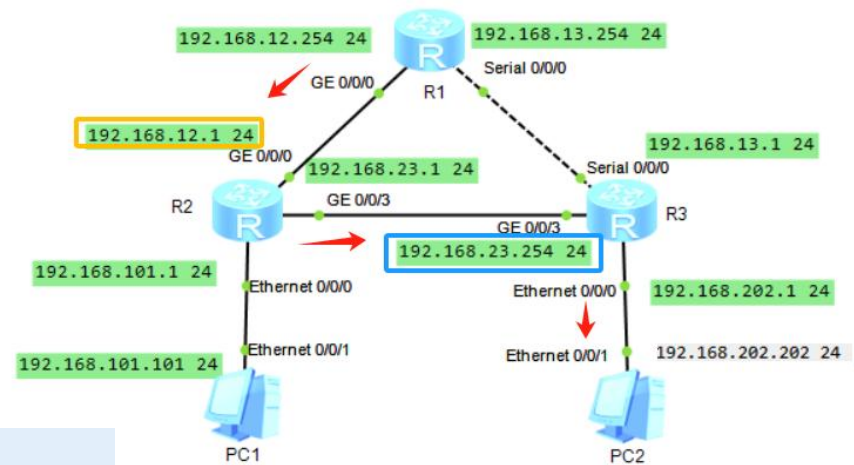
OSPF(4)

Q. How to calculate the “cost” in OSPF ?

link cost

R1->R3->PC2 : 1562 + 1 = 1563

R1->R2->R3->PC2 : 1+1+1 = 3



```

R1
[Huawei]dis ospf routing

  OSPF Process 1 with Router ID 192.168.12.254
    Routing Tables

  Routing for Network
  Destination      Cost  Type      NextHop      AdvRouter      Area
  192.168.12.0/24  1     Transit   192.168.12.254 192.168.12.254 0.0.0.0
  192.168.13.0/24  1562  Stub      192.168.13.254 192.168.12.254 0.0.0.0
  192.168.23.0/24  2     Transit   192.168.12.1  192.168.101.1  0.0.0.0
  192.168.101.0/24  2     Stub      192.168.12.1  192.168.101.1  0.0.0.0
  192.168.202.0/24  3     Stub      192.168.12.1  192.168.202.1  0.0.0.0

  Total Nets: 5
  Intra Area: 5  Inter Area: 0  ASE: 0  NSSA: 0

[Huawei]
  
```

The route with lowest link cost is the final choice for the Router, so R1->R2->R3->PC2 is the final choice.

```

R1
[Huawei]dis ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 10      Routes : 10

Destination/Mask    Proto    Pre  Cost      Flags NextHop      Interface
-----
  127.0.0.0/8       Direct   0    0          D  127.0.0.1      InLoopBack0
  127.0.0.1/32       Direct   0    0          D  127.0.0.1      InLoopBack0
  192.168.12.0/24    Direct   0    0          D  192.168.12.254 GigabitEthernet
0/0/0
  192.168.12.254/32  Direct   0    0          D  127.0.0.1      GigabitEthernet
0/0/0
  192.168.13.0/24    Direct   0    0          D  192.168.13.254 Serial0/0/0
  192.168.13.1/32    Direct   0    0          D  192.168.13.1   Serial0/0/0
  192.168.13.254/32  Direct   0    0          D  127.0.0.1      Serial0/0/0
  192.168.23.0/24    OSPF     10    2          D  192.168.12.1   GigabitEthernet
0/0/0
  192.168.101.0/24   OSPF     10    2          D  192.168.12.1   GigabitEthernet
0/0/0
  192.168.202.0/24   OSPF     10    3          D  192.168.12.1   GigabitEthernet
0/0/0
  
```

Part B. ARP for IPv4 vs NS&NA for IPv6

- At the transport layer, network layer, and link layer, it is necessary to use **source** and **destination** to mark the identity of the communicating parties in the current transmission direction.
- The **transport** layer uses **port** numbers, the **network** layer uses **IP addresses**, and the **link** layer uses **physical addresses**(MAC address).

Q. How to know the identity of the communication partner on different protocol layers (port, IP address, MAC address)?

```
> Frame 1: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF {7E0D759F-11
> Ethernet II, Src: IntelCor d2:20:30 (04:ed:33:d2:20:30), Dst: MaipuCom 42:fe:07 (cc:d8:1f:42:fe:07)
> Internet Protocol Version 4, Src: 192.168.10.123, Dst: 8.8.8.8
> Transmission Control Protocol, Src Port: 49804, Dst Port: 443, Seq: 0, Len: 0
```

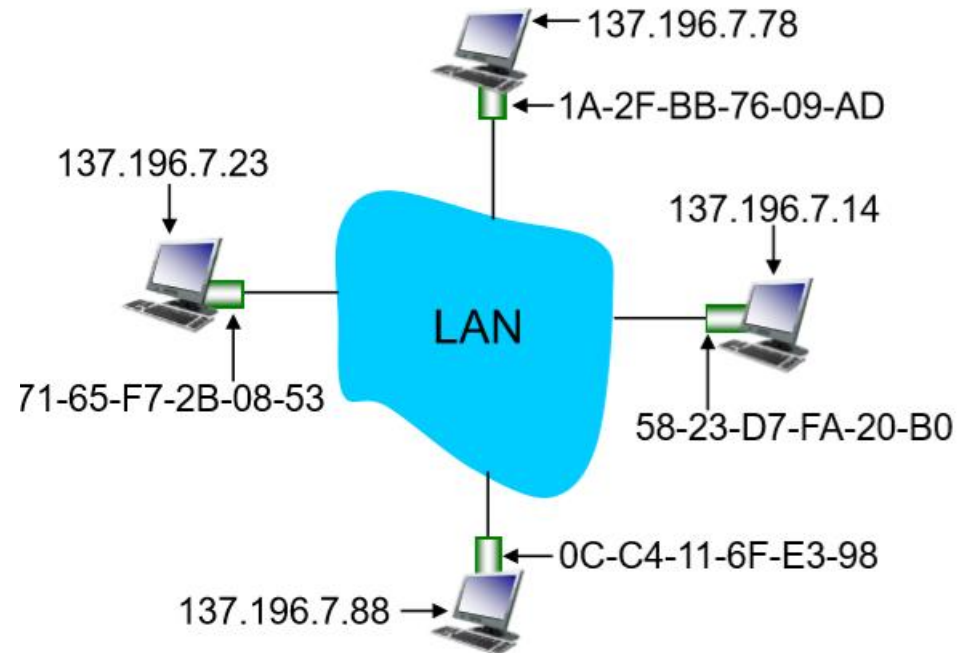
icmpv6					
No.	Time	Source	Protocol	Destination	Info
2	9.688000	2025::101	ICMPv6	ff15::1	Echo (ping) request id=0x4619, seq=1,

```
> Frame 2: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface -, id 0
> Ethernet II, Src: HuaweiTe_fd:64:76 (54:89:98:fd:64:76), Dst: IPv6mcast_01 (33:33:00:00:00:01)
> Internet Protocol Version 6, Src: 2025::101, Dst: ff15::1
> Internet Control Message Protocol v6
```

MAC address

- MAC (or LAN or physical or Ethernet) address:
 - function: *used 'locally' to get frame from one interface to another physically-connected interface (same network, in IP-addressing sense)*
 - 48 bit MAC address (for most LANs) burned in NIC ROM, also sometimes software settable
 - e.g.: 1A-2F-BB-76-09-AD

hexadecimal (base 16) notation
(each "numeral" represents 4 bits)



ARP(Address Resolution Protocol)

```
23 3.409057 Micro-St_b3:5c:39 Broadcast ARP Who has 172.18.130.25? Tell 172.18.130.27
24 3.409348 Micro-St_b0:d9:cd Micro-St_b3:5c... ARP 172.18.130.25 is at 44:8a:5b:b0:d9:cd
<
> Frame 23: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
> Ethernet II, Src: Micro-St_b3:5c:39 (44:8a:5b:b3:5c:39), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
v Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Micro-St_b3:5c:39 (44:8a:5b:b3:5c:39)
  Sender IP address: 172.18.130.27 (172.18.130.27)
  Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
  Target IP address: 172.18.130.25 (172.18.130.25)
```

1. using “arp -d” to clear the mac-address table on PC.
2. “ping” an reachable IP.
3. use “arp” in WireShark to display ARP frame.
4. use “eth.addr == *.*.*.*.*.*.*” in WireShark to filter MAC addresses.

```
24 3.409348 Micro-St_b0:d9:cd Micro-St_b3:5c... ARP 172.18.130.25 is at 44:8a:5b:b0:d9:cd
<
> Frame 24: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
> Ethernet II, Src: Micro-St_b0:d9:cd (44:8a:5b:b0:d9:cd), Dst: Micro-St_b3:5c:39 (44:8a:5b:b3:5c:39)
v Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: Micro-St_b0:d9:cd (44:8a:5b:b0:d9:cd)
  Sender IP address: 172.18.130.25 (172.18.130.25)
  Target MAC address: Micro-St_b3:5c:39 (44:8a:5b:b3:5c:39)
  Target IP address: 172.18.130.27 (172.18.130.27)
```

```
[H3C]display mac-address
MAC Address      VLAN ID  State      Port/Nickname  Aging
448a-5bb3-5c39   1        Learned    GE1/0/23       Y
448a-5bb3-5f55   1        Learned    GE1/0/1        Y
[H3C]
```

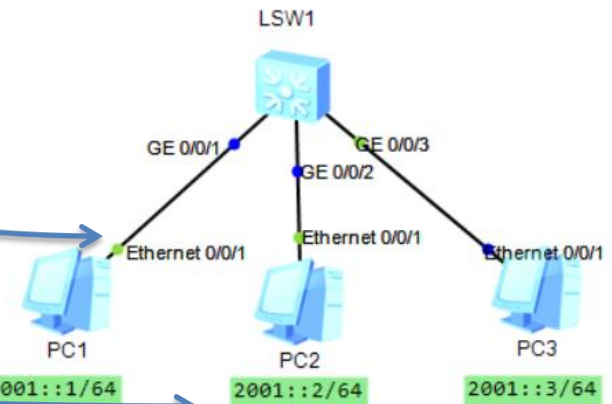
Neighbor Solicitation in ICMPv6

```
101 ... 2001::1 ICMPv6 ff02::1:ff00:2 Neighbor Solicitation for 2001::2 from 54:89:98:d7:2f:86
> Frame 101: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface -, id 0
> Ethernet II, Src: HuaweiTe_d7:2f:86 (54:89:98:d7:2f:86), Dst: IPv6mcast_ff:00:00:02 (33:33:ff:00:00:02)
> Internet Protocol Version 6, Src: 2001::1, Dst: ff02::1:ff00:2
    0110 .... = Version: 6
    > .... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
    .... 0000 0000 0000 0000 0000 = Flow Label: 0x000000
    Payload Length: 32
    Next Header: ICMPv6 (58)
    Hop Limit: 255
    Source Address: 2001::1
    Destination Address: ff02::1:ff00:2
    [Source Teredo Server IPv4: 0.0.0.0]
    [Source Teredo Port: 65535]
    [Source Teredo Client IPv4: 255.255.255.254]
> Internet Control Message Protocol v6
    Type: Neighbor Solicitation (135)
    Code: 0
    Checksum: 0x1cb1 [correct]
    [Checksum Status: Good]
    Reserved: 00000000
    Target Address: 2001::2
    > ICMPv6 Option (Source link-layer address : 54:89:98:d7:2f:86)
        Type: Source link-layer address (1)
        Length: 1 (8 bytes)
        Link-layer address: HuaweiTe_d7:2f:86 (54:89:98:d7:2f:86)
```

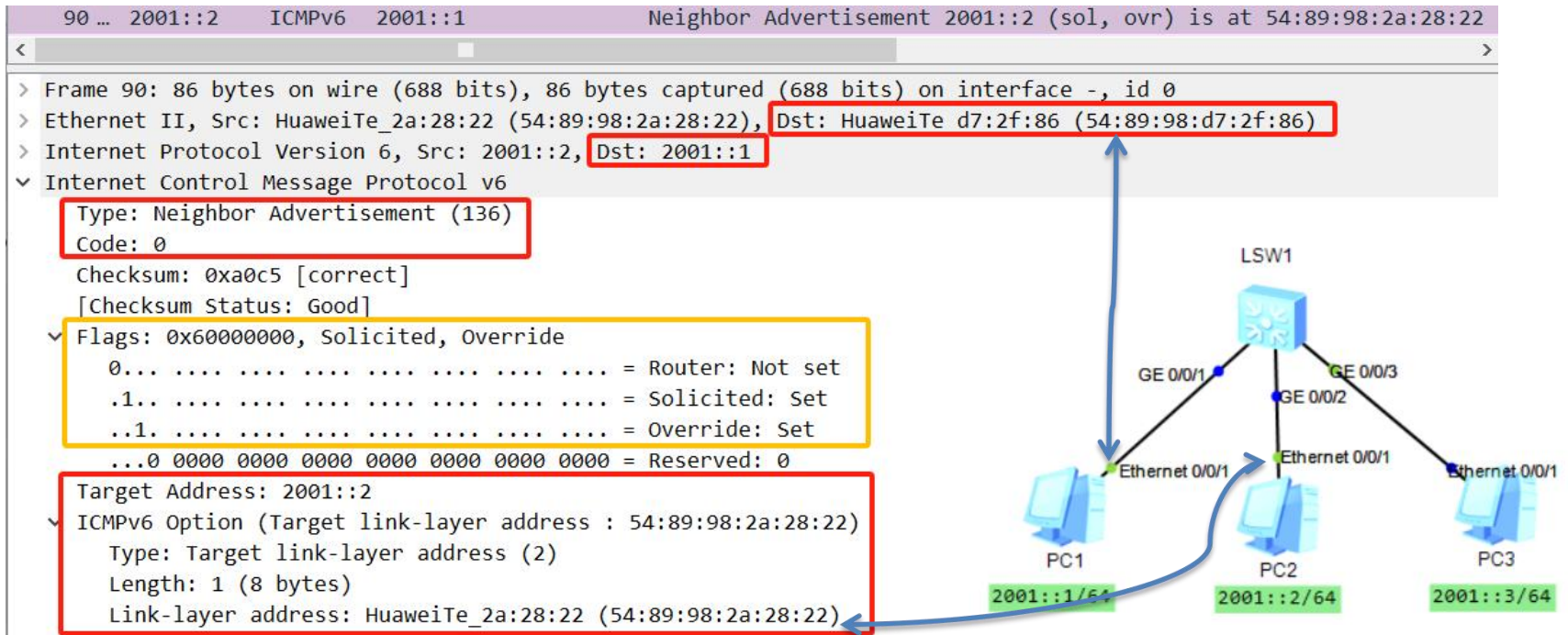
➤ Nodes send **Neighbor Solicitations (NS)** to **request the link-layer address of a target node** while also providing their own link-layer address to the target.

➤ **Neighbor Solicitations** are **multicast** when the node needs to resolve an address and **unicast** when the node seeks to verify the reachability of a neighbor.

invoke “ping” on PC1 to PC2.



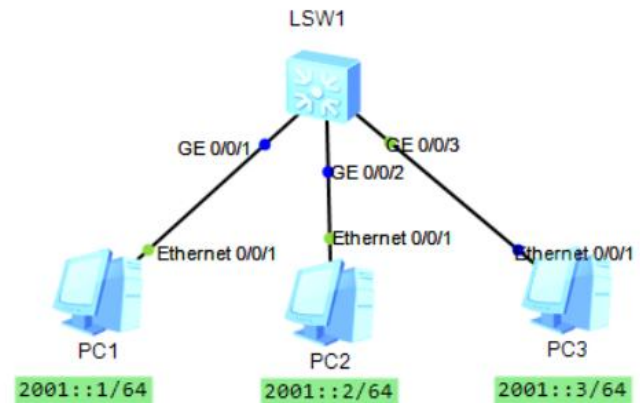
Neighbor Advertisements in ICMPv6(1)



- Router flag. When set, the R-bit indicates that the sender is a router.
- Solicited flag. When set, the S-bit indicates that **the advertisement was sent in response to a Neighbor Solicitation from the Destination address.**
- Override flag. When set, the O-bit indicates that **the advertisement should override an existing cache entry and update the cached link-layer address.**

Neighbor Advertisements in ICMPv6(2)

```
90 ... 2001::2 ICMPv6 2001::1 Neighbor Advertisement 2001::2 (sol, ovr) is at 54:89:98:2a:28:22
> Frame 90: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface -, id 0
> Ethernet II, Src: HuaweiTe_2a:28:22 (54:89:98:2a:28:22), Dst: HuaweiTe d7:2f:86 (54:89:98:d7:2f:86)
> Internet Protocol Version 6, Src: 2001::2, Dst: 2001::1
> Internet Control Message Protocol v6
  Type: Neighbor Advertisement (136)
  Code: 0
  Checksum: 0xa0c5 [correct]
  [Checksum Status: Good]
  Flags: 0x60000000, Solicited, Override
    0... .. = Router: Not set
    .1... .. = Solicited: Set
    ..1... .. = Override: Set
    ...0 0000 0000 0000 0000 0000 0000 0000 = Reserved: 0
  Target Address: 2001::2
  ICMPv6 Option (Target link-layer address : 54:89:98:2a:28:22)
    Type: Target link-layer address (2)
    Length: 1 (8 bytes)
    Link-layer address: HuaweiTe_2a:28:22 (54:89:98:2a:28:22)
```



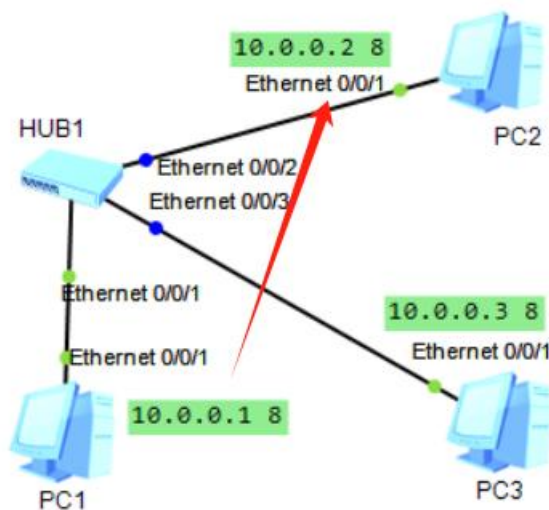
- A node sends **Neighbor Advertisements** in response to **Neighbor Solicitations** and sends unsolicited Neighbor Advertisements in order to (unreliably) propagate new information quickly.
- **Target link-layer address.** The link-layer address for the target, i.e., the sender of the advertisement.

Part C. Device

Hub(Layer 1) broadcast(1)

Hub broadcasts the package while not check the destination address of it.

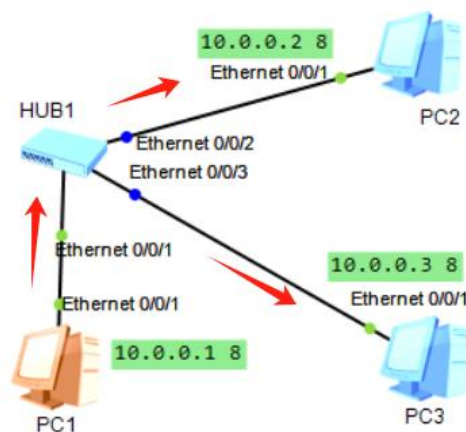
Demo1. Build a simple network topology, with 3 PCs and a Hub. Complete the basic configuration of the PCs as shown in the figure. Capture packets on the Ethernet 0/0/2 and Ethernet 0/0/3, then initiate “ping” test from PC1 to PC2.



```
PC1
Basic Config Command MCPacket UdpPacket Console
PC>ipconfig
Link local IPv6 address.....: fe80::5689:98ff:fe56:27ca
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.0.0.1
Subnet mask.....: 255.0.0.0
Gateway.....: 0.0.0.0
Physical address.....: 54-89-98-56-27-CA
DNS server.....:
PC>ping 10.0.0.2
Ping 10.0.0.2: 32 data bytes, Press Ctrl_C to break
From 10.0.0.2: bytes=32 seq=1 ttl=128 time=31 ms
From 10.0.0.2: bytes=32 seq=2 ttl=128 time=31 ms
```

Hub(Layer 1) broadcast(2)

Capture packets on the Ethernet 0/0/2 and Ethernet 0/0/3, then initiate “ping” test from PC1 to PC2.



Q1. Which device sends ARP request? while HUB1 receives the ARP request from its interface X, would it broadcast the ARP request to the interface X?

Q2. Which device sends ARP reply? would PC3 receive the ARP reply ?

Q3. What's the ARP-Table on PC1, PC2 and PC3 after the “ping” test from PC1 to PC2 ?

Source	Protocol	Destination	Length	Info	Source	Protocol	Destination	Length	Info
HuaweiTe_56:2...	ARP	Broadcast	60	Who has 10.0.0.2? Tell 10.0.0.1	HuaweiTe_...	ARP	Broadcast	60	Who has 10.0.0.2? Tell
HuaweiTe_5a:4...	ARP	HuaweiTe...	60	10.0.0.2 is at 54:89:98:5a:45:72	HuaweiTe_...	ARP	HuaweiTe...	60	10.0.0.2 is at 54:89:9
10.0.0.1	ICMP	10.0.0.2	74	Echo (ping) request id=0x7f01, seq=1/256, ttl=	10.0.0.1	ICMP	10.0.0.2	74	Echo (ping) request i
10.0.0.2	ICMP	10.0.0.1	74	Echo (ping) reply id=0x7f01, seq=1/256, ttl=	10.0.0.2	ICMP	10.0.0.1	74	Echo (ping) reply i
10.0.0.1	ICMP	10.0.0.2	74	Echo (ping) request id=0x8001, seq=2/512, ttl=	10.0.0.1	ICMP	10.0.0.2	74	Echo (ping) request i
10.0.0.2	ICMP	10.0.0.1	74	Echo (ping) reply id=0x8001, seq=2/512, ttl=	10.0.0.2	ICMP	10.0.0.1	74	Echo (ping) reply i
10.0.0.1	ICMP	10.0.0.2	74	Echo (ping) request id=0x8101, seq=3/768, ttl=	10.0.0.1	ICMP	10.0.0.2	74	Echo (ping) request i
10.0.0.2	ICMP	10.0.0.1	74	Echo (ping) reply id=0x8101, seq=3/768, ttl=	10.0.0.2	ICMP	10.0.0.1	74	Echo (ping) reply i
10.0.0.1	ICMP	10.0.0.2	74	Echo (ping) request id=0x8201, seq=4/1024, tt=	10.0.0.1	ICMP	10.0.0.2	74	Echo (ping) request i
10.0.0.2	ICMP	10.0.0.1	74	Echo (ping) reply id=0x8201, seq=4/1024, tt=	10.0.0.2	ICMP	10.0.0.1	74	Echo (ping) reply i
10.0.0.1	ICMP	10.0.0.2	74	Echo (ping) request id=0x8301, seq=5/1280, tt=	10.0.0.1	ICMP	10.0.0.2	74	Echo (ping) request i
10.0.0.2	ICMP	10.0.0.1	74	Echo (ping) reply id=0x8301, seq=5/1280, tt=	10.0.0.2	ICMP	10.0.0.1	74	Echo (ping) reply i

packets captured on Eth0/0/2

packets captured on Eth0/0/3

Hub(Layer 1) broadcast(3)

```
PC1
Basic Config Command MCPacket UdpPacket Console
PC>ping 10.0.0.2

Ping 10.0.0.2: 32 data bytes, Press Ctrl_C to break
From 10.0.0.2: bytes=32 seq=1 ttl=128 time=32 ms
From 10.0.0.2: bytes=32 seq=2 ttl=128 time=31 ms
From 10.0.0.2: bytes=32 seq=3 ttl=128 time=31 ms
From 10.0.0.2: bytes=32 seq=4 ttl=128 time=32 ms
From 10.0.0.2: bytes=32 seq=5 ttl=128 time=31 ms

--- 10.0.0.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/31/32 ms

PC>arp -a

Internet Address      Physical Address      Type
10.0.0.2              54-89-98-5A-45-72    dynamic
```

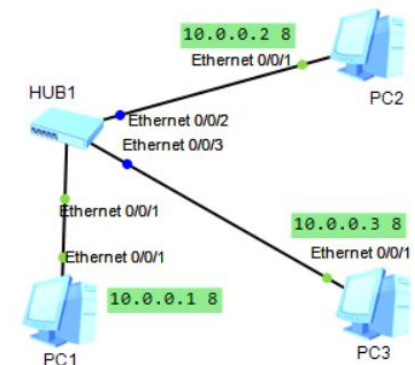
```
PC2
Basic Config Command MCPacket UdpPacket Console
PC>arp -a

Internet Address      Physical Address      Type
10.0.0.1              54-89-98-56-27-CA    dynamic
```

```
PC3
Basic Config Command MCPacket UdpPacket Console
PC>arp -a

Internet Address      Physical Address      Type
10.0.0.2              54-89-98-5A-45-72    dynamic
10.0.0.1              54-89-98-56-27-CA    dynamic
```

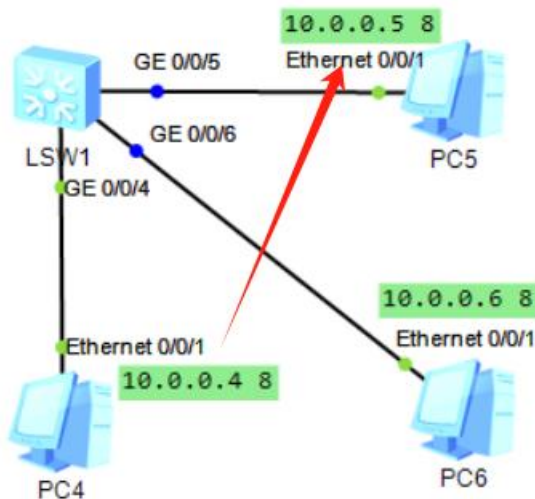
- PC1 learns the MAC address of PC2 by ARP reply from PC2
- PC2 learns the MAC address of PC1 by ARP request from PC1
- PC3 learns the MAC address of PC1 by ARP reply from PC2, learns the MAC address of PC1 by ARP request from PC1.



Switch(Layer 2) - Mac address table(1)

- **Switch** can **learn** from the received package: to gain its source mac address and the interface id to make a **Mac-address Table**.
- **Switch** use the **Mac-address Table** to forward the package on Layer2.

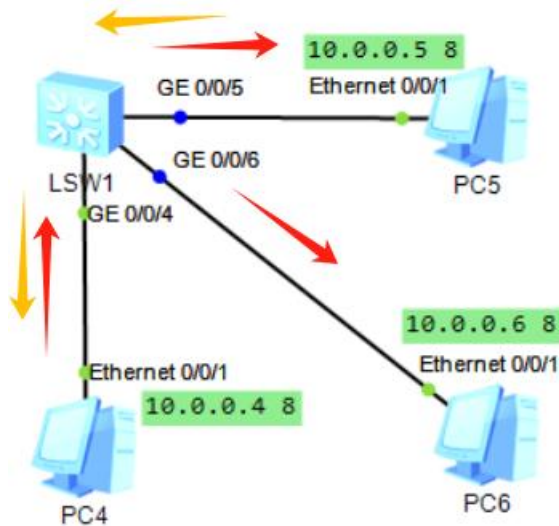
Demo2. Build a simple network topology, with 3 PCs and a Switch. Complete the basic configuration of the PCs as shown in the figure. Capture packets on the Ethernet 0/0/5 and Ethernet 0/0/6, then initiate “ping” test from PC4 to PC5.



```
PC4
Basic Config  Command  MCPacket  UDPacket  Console
PC>ipconfig
Link local IPv6 address.....: fe80::5689:98ff:fe0a:2252
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.0.0.4
Subnet mask.....: 255.0.0.0
Gateway.....: 0.0.0.0
Physical address.....: 54-89-98-0A-22-52
DNS server.....:
PC>ping 10.0.0.5
Ping 10.0.0.5: 32 data bytes, Press Ctrl_C to break
From 10.0.0.5: bytes=32 seq=1 ttl=128 time=47 ms
From 10.0.0.5: bytes=32 seq=2 ttl=128 time=47 ms
From 10.0.0.5: bytes=32 seq=3 ttl=128 time=47 ms
From 10.0.0.5: bytes=32 seq=4 ttl=128 time=47 ms
From 10.0.0.5: bytes=32 seq=5 ttl=128 time=46 ms
```

Switch - Mac address table(2)

Capture packets on the GE 0/0/5 and GE 0/0/6, then initiate “ping” test from PC4 to PC5.



Q1. While ARP request reaches the Switch(LSW1), what would the Switch do? broadcast it or send it to PC5 ? How about ARP reply reaches the Switch(LSW1)?

Q2. While ICMP request reaches the Switch(LSW1), what would the Switch do? broadcast it or send it to PC5 ? How about ICMP reply reaches the Switch(LSW1)?

Q3. What's the ARP-Table on PC4, PC5 and PC6 after the “ping” test from PC4 to PC6 ? What's the MAC-Address table on the Switch(LSW1) ?

packets captured on GE0/0/5 of Switch

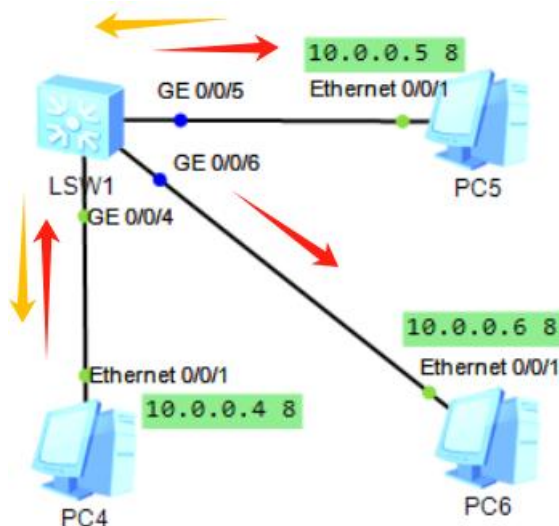
No.	Time	Source	Protocol	Destination	Length	Info
...	32.703000	HuaweiTe_0a:2...	ARP	Broadcast	60	Who has 10.0.0.5? Tell 10.0.0.4
...	32.719000	HuaweiTe_f0:1...	ARP	HuaweiTe...	60	10.0.0.5 is at 54:89:98:f0:18:25
...	32.781000	10.0.0.4	ICMP	10.0.0.5	74	Echo (ping) request id=0xa90c, seq=1/256, ttl=128
...	32.797000	10.0.0.5	ICMP	10.0.0.4	74	Echo (ping) reply id=0xa90c, seq=1/256, ttl=128
...	33.844000	10.0.0.4	ICMP	10.0.0.5	74	Echo (ping) request id=0xaa0c, seq=2/512, ttl=128
...	33.844000	10.0.0.5	ICMP	10.0.0.4	74	Echo (ping) reply id=0xaa0c, seq=2/512, ttl=128
...	34.891000	10.0.0.4	ICMP	10.0.0.5	74	Echo (ping) request id=0xab0c, seq=3/768, ttl=128
...	34.906000	10.0.0.5	ICMP	10.0.0.4	74	Echo (ping) reply id=0xab0c, seq=3/768, ttl=128
...	35.953000	10.0.0.4	ICMP	10.0.0.5	74	Echo (ping) request id=0xac0c, seq=4/1024, ttl=128
...	35.953000	10.0.0.5	ICMP	10.0.0.4	74	Echo (ping) reply id=0xac0c, seq=4/1024, ttl=128
...	37.016000	10.0.0.4	ICMP	10.0.0.5	74	Echo (ping) request id=0xad0c, seq=5/1280, ttl=128
...	37.031000	10.0.0.5	ICMP	10.0.0.4	74	Echo (ping) reply id=0xad0c, seq=5/1280, ttl=128

packets captured on GE0/0/6 of Switch

No.	Time	Source	Protocol	Destination	Length	Info
...	23.703000	HuaweiTe_0a:2...	ARP	Broadcast	60	Who has 10.0.0.5? Tell 10.0.0.4

Switch - Mac address table(3)

- Switch can **learn** from the received package: to gain its source mac address and the interface id to make a **Mac-address Table**.
- Switch use the Mac-address Table to forward the package on Layer2.



```
[Huawei]dis mac-address
MAC address table of slot 0:
-----
MAC Address      VLAN/      PEVLAN CEVLAN Port      Type      LSP/LSR-ID
                  VSI/SI
-----
5489-98f0-1825  1          -      -      GE0/0/5      dynamic  0/-
5489-980a-2252  1          -      -      GE0/0/4      dynamic  0/-
-----
Total matching items on slot 0 displayed = 2
```

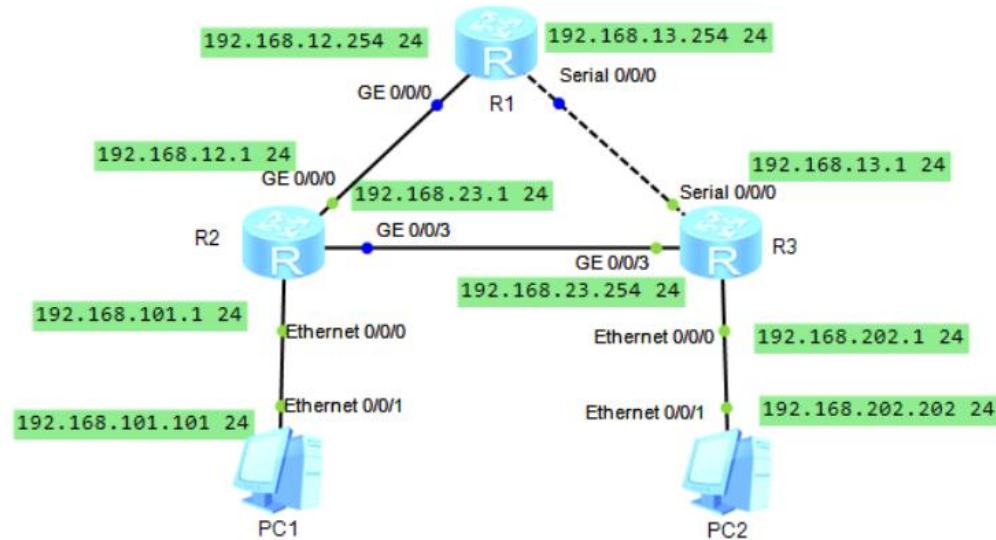
In this demo, Switch (LSW1) learns the MAC address of PC4 from the interface GE0/0/4 while received the ARP request packet. Switch(LSW1) learns the MAC address of PC5 from the interface GE0/0/5 while received the ARP reply packet.

Q. If the MAC address of PC5 is added by “add -s” on PC4 before PC4 invoke “ping” test to PC5, what would the Switch (LSW1) do after receiving the ICMP request from PC4?

Practice 12.1(1)

Build and set the IPv4 network as below topology, do the following test on eNSP

- Configure the RIPv2 protocol on the router based on the topology in the figure on the right.
- After the RIPv2 protocol is configured, Each node in the network should be reachable at the network layer.
 - ✓ using “ping” for testing.
 - ✓ using “dis ip routing-table” to show RIPv2 works, and there should be no static route item or the route item generated by other routing protocol.
- Compare the behavior of RIPv2 with OSPF(which is demonstrated on page 9-12 in this slides) and complete the table on the next page.



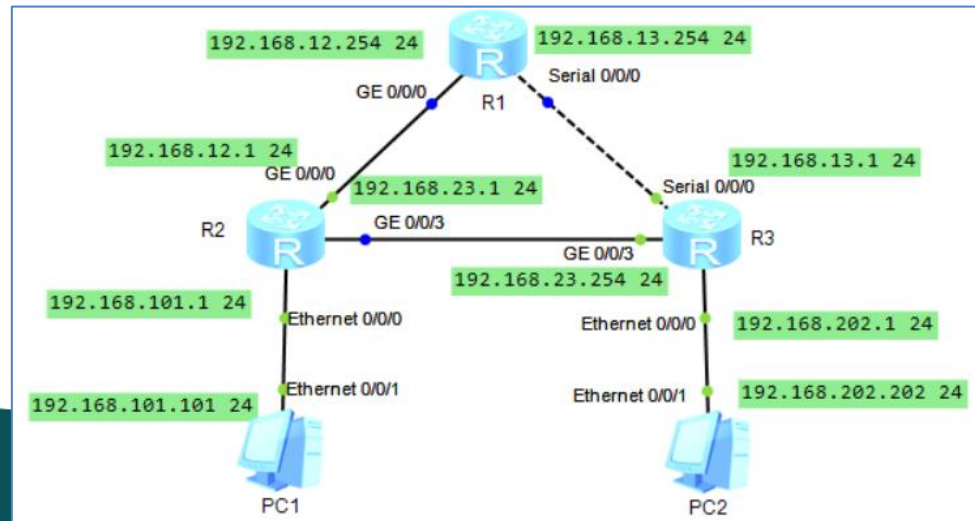
Practice 12.1(2)

List the key information of non direct routing in R1 router in the below table based on your experiment for comparison on RIPv2 and OSPF.

R1	destination/Mask	proto	pre	cost	nexthop	interface
OSPF	192.168.23.0/24					
RIPv2	192.168.23.0/24					
OSPF	192.168.101.0/24					
RIPv2	192.168.101.0/24					
OSPF	192.168.202.0/24					
RIPv2	192.168.202.0/24					

TIPS:

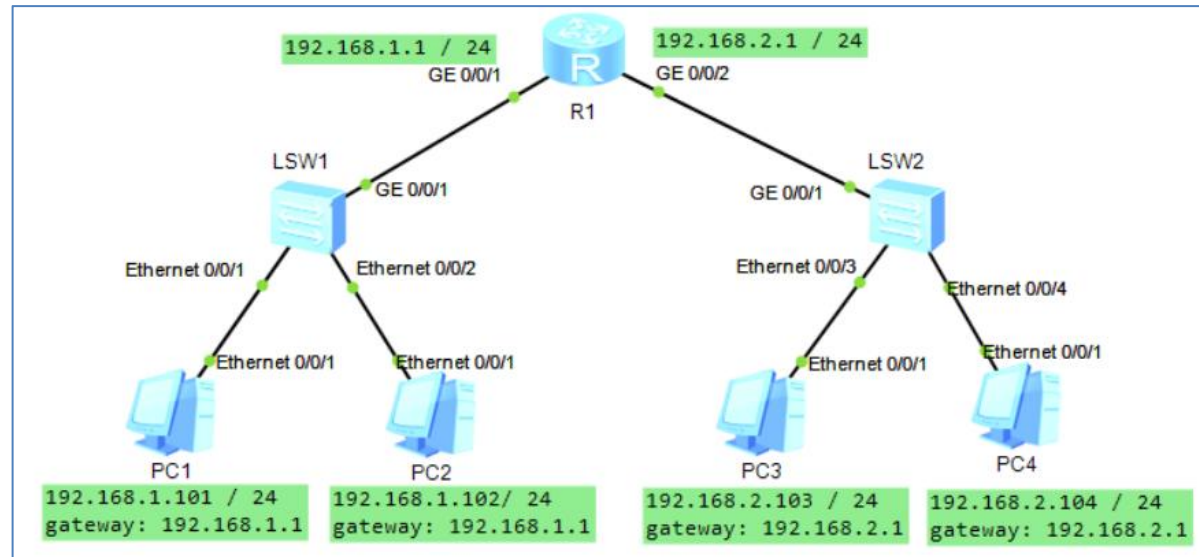
1. The relevant information about OSPF can be found on page 9-12 of the lab courseware.
2. Use the “dis ip routing-table” on the router to view key information of the routing table



Practice 12.2(1)

Build and set the IPv4 network as below topology, do the following test on eNSP

- **Clear** the ARP table on the PC and the MAC address table on the Switches.
 - using “arp -d” on PC3 to clear its arp-table
 - using “undo mac-address” on LSW1 and LSW2 to clear its mac-address table.
- DO the “ping” tests
 - Invoke “ping” on PC3 to reach PC4.
 - Invoke “ping” on PC3 to reach PC1.

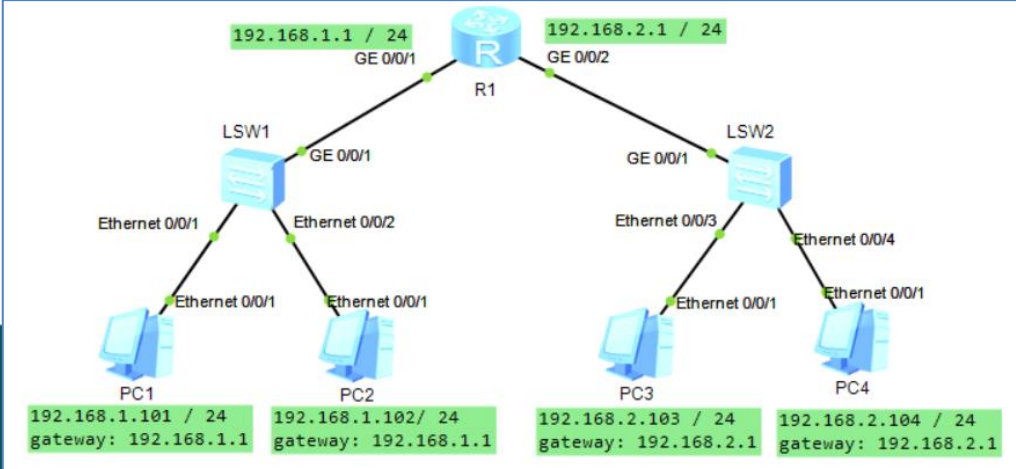


Observe the ARP protocol messages, the ARP table on the PC, and the MAC address table on the switches, Complete the table on the next page according to your experiments.

Practice 12.2(2)

The arp request referred to in the following text specifically refers to the one sent by PC3	invoke“ping”on PC3 to reach PC4	invoke“ping”on PC3 to reach PC1
sender IP in the ARP request		
target IP in the ARP request		
While the ARP request reach to LSW2, list interfaces on LSW2 on which the ARP request was forwarded from. (Eth0/0/3, Eth0/0/4, Ge0/0/1)		
Does the ARP request reach to the router?(Y/N)		
If the ARP request reach to the router, what does the router do after receiving the ARP request? (forward/drop/reply)		
Which network nodes' MAC addresses would been learnt by LSW2 in this testing? (PC1, PC2, PC3, PC4, which interface of R1, which interface of LSW1, which interface of LSW2)		

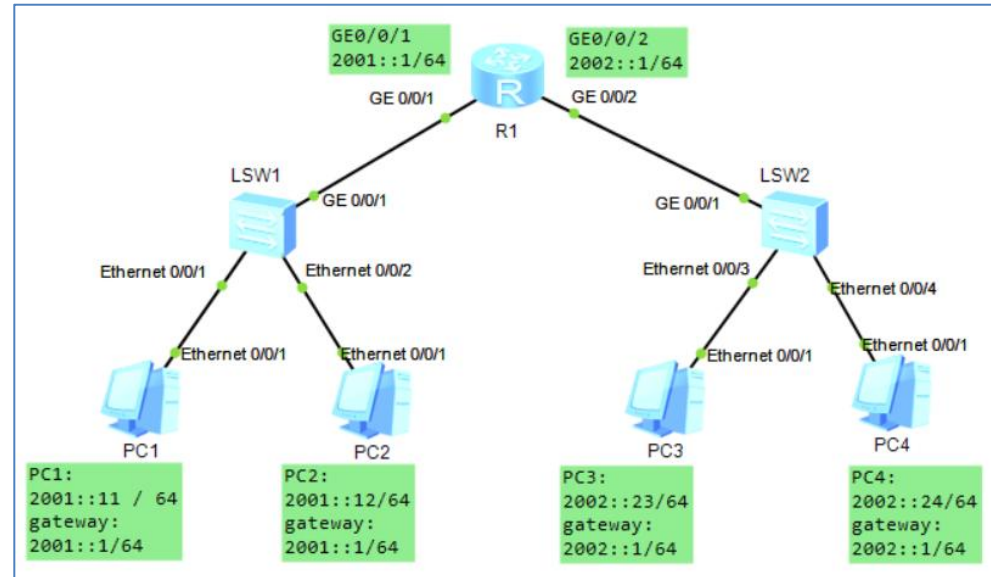
NOTES: Do Clear the ARP table on the PC and the MAC address table on the Switches before each “ping” test.
Refer to the previous page for specific steps.



Practice 12.3(1)

Build and set the IPv6 network as below topology, do the following test on eNSP

- using “**arp -d**” on **PC1** to clear its arp-table
- using “**undo mac-address**” on **LSW1, LSW2** and **R1** to clear its mac-address table.
- Invoke “**ping 2002::23**” on **PC1** to reach **PC3**
- Observe:
 - the Neighbor Solicitation and Neighbor Advertisement protocol messages
 - the ARP table on the PCs
 - the MAC address table on the switches and the router
- Answer the questions on the next page according to your experiments



TIPS: “dis int <interface name>” in system view could be used to show the details about the interface on both switch and router.

Practice 12.3(2)

invoke“ping”on PC1 to reach PC3

Has the router received the NS message with “target address” 2001:: 1 ?

Who(pc1, pc2, pc3, pc4) sent it?

Is the destination of this message a unicast address or a multicast address?

Which device and interface the “Source link layer address” is for?

Has this device received the NA message?

Which device's MAC address for which interface is the target is the “Target link-layer address” in the NA message?

Has the router received the NS message with “target address” 2002:: 1 ?

Who(pc1, pc2, pc3, pc4) sent it?

Is the destination of this message a unicast address or a multicast address ?

Which device and interface the “Source link layer address” is for?

Has this device received the NA message?

Which device's MAC address for which interface is the target is the “Target link-layer address” in the NA message?

Which network nodes' MAC addresses would be learnt by LSW1 in this testing?

(PC1, PC2, PC3, PC4, which interface of R1, which interface of LSW1, which interface of LSW2)

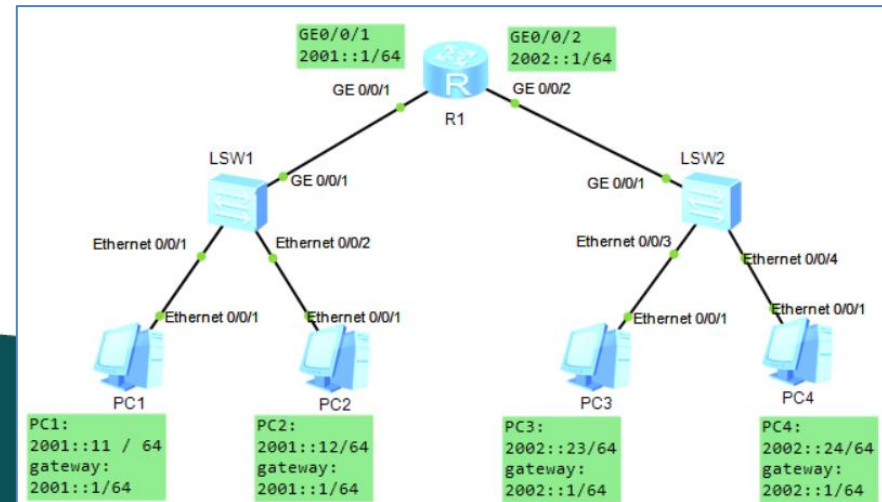
Which network nodes' MAC addresses would be learnt by LSW2 in this testing?

(PC1, PC2, PC3, PC4, which interface of R1, which interface of LSW1, which interface of LSW2)

TIPS:

<https://datatracker.ietf.org/doc/html/rfc2461#section-4.2>

<https://datatracker.ietf.org/doc/html/rfc2461#section-4.3>



TIPS:How to check the MAC address of the device interface

using “dis int <interface name>” to show the details about the interface on both switch and router

```
LSW2 LSW1 R1
<Huawei>dis int Ethernet 0/0/3
Ethernet0/0/3 current state : UP
Line protocol current state : UP
Description:
Switch Port, PVID : 1, TPID : 8100(Hex), The Maximum Frame Length is 9216
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 4c1f-cc0f-028b
Last physical up time : 2025-12-01 16:12:51 UTC-08:00
Last physical down time : 2025-12-01 16:12:49 UTC-08:00
```

```
LSW2 LSW1 R1
<Huawei>dis int gi 0/0/2
GigabitEthernet0/0/2 current state : UP
Line protocol current state : UP
Last line protocol up time : 2025-12-01 16:12:50 UTC-08:00
Description:
Route Port,The Maximum Transmit Unit is 1500
Internet Address is 192.168.2.1/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 5489-98af-60da
```

PC3

Basic Config Command MCPacket UdpPacket Console

Host Name:

MAC Address: